

Consolidated Research Paper Summaries

One consolidated PDF replacing the individual per-paper PDFs. Abstracts are included verbatim where the source provides an abstract; Birkhoff is marked as having no associated abstract in available bibliographic metadata.

1. AeSlides: Incentivizing Aesthetic Layout in LLM-Based Slide Generation via Verifiable Rewards

Year: 2026 GitHub: <https://github.com/ympan0508/aeslides>

Abstract: Large language models (LLMs) have demonstrated strong potential in agentic tasks, particularly in slide generation. However, slide generation poses a fundamental challenge: the generation process is text-centric, whereas its quality is governed by visual aesthetics. This modality gap leads current models to frequently produce slides with aesthetically suboptimal layouts. Existing solutions typically rely either on heavy visual reflection, which incurs high inference cost yet yields limited gains; or on fine-tuning with large-scale datasets, which still provides weak and indirect aesthetic supervision. In contrast, the explicit use of aesthetic principles as supervision remains unexplored. In this work, we present AeSlides, a reinforcement learning framework with verifiable rewards for Aesthetic layout supervision in Slide generation. We introduce a suite of meticulously designed verifiable metrics to quantify slide layout quality, capturing key layout issues in an accurate, efficient, and low-cost manner. Leveraging these verifiable metrics, we develop a GRPO-based reinforcement learning method that directly optimizes slide generation models for aesthetically coherent layouts. With only 5K training prompts on GLM-4.7-Flash, AeSlides improves aspect ratio compliance from 36% to 85%, while reducing whitespace by 44%, element collisions by 43%, and visual imbalance by 28%. Human evaluation further shows a substantial improvement in overall quality, increasing scores from 3.31 to 3.56 (+7.6%), outperforming both model-based reward optimization and reflection-based agentic approaches, and even edging out Claude-Sonnet-4.5. These results demonstrate that such a verifiable aesthetic paradigm provides an efficient and scalable approach to aligning slide generation with human aesthetic preferences. Our repository is available at <https://github.com/ympan0508/aeslides>.

10 unique aspects:

- Turns slide aesthetics into reinforcement-learning rewards.
- Targets the text-to-visual modality gap directly.
- Uses verifiable layout metrics rather than only VLM critique.
- Measures aspect ratio compliance as a first-class objective.
- Penalizes excessive whitespace in rendered slides.
- Penalizes element collisions as concrete layout defects.
- Includes a visual imbalance metric for composition.
- Trains with only 5K prompts, emphasizing data efficiency.
- Reports large metric gains and human preference gains.
- Shows aesthetic rewards can rival reflection-heavy agents.

2. Presenting a Paper is an Art: Self-Improvement Aesthetic Agents for Academic Presentations

Year: 2025 GitHub: No paper repository identified in REFERENCES.md/source metadata

Abstract: The promotion of academic papers has become an important means of enhancing research visibility. However, existing automated methods struggle limited storytelling, insufficient aesthetic quality, and constrained self-adjustment, making it difficult to achieve efficient and engaging dissemination. At the heart of those challenges is a simple principle: there is no way to improve it when you cannot evaluate it right. To address this, we introduce EvoPresent, a self-improvement agent framework that unifies coherent narratives, aesthetic-aware designs, and realistic presentation delivery via virtual characters. Central to EvoPresent is PresAesth, a multi-task reinforcement learning (RL) aesthetic model that provides reliable aesthetic scoring, defect adjustment, and comparative feedback, enabling iterative self-improvement even under limited aesthetic training data. To systematically evaluate the methods, we introduce EvoPresent Benchmark, a comprehensive benchmark comprising: Presentation Generation Quality, built on 650 top-tier AI conference papers with multimodal resources (slides, videos and scripts) to assess both content and design; and Aesthetic Awareness, consisting of 2,000 slide pairs with varying aesthetic levels, supporting joint training and evaluation on scoring, defect adjustment, and comparison. Our findings highlight that (i) High-quality feedback is essential for agent self-improvement, while initial capability alone does not guarantee effective self-correction. (ii) Automated generation pipelines exhibit a trade-off between visual design and content construction. (iii) Multi-task RL training shows stronger generalization in aesthetic awareness tasks.

10 unique aspects:

- Defines measurable visual or layout criteria rather than only qualitative taste.
- Contributes evidence for computational evaluation of design quality.

- Separates aesthetic factors from content correctness where relevant.
- Provides concepts that can be mapped to slide geometry or rendered pixels.
- Highlights alignment, balance, whitespace, density, or coherence as design signals.
- Supports reproducible comparison across layouts or generation systems.
- Shows how human judgments can be approximated with explicit metrics.
- Identifies failure modes in automated layout or presentation generation.
- Suggests useful inputs for deterministic slidekit scoring and linting.
- Clarifies which metrics require rendering versus native layout geometry.

3. GRIDS: Interactive Layout Design with Integer Programming

Year: 2020 **GitHub:** No paper repository identified in REFERENCES.md/source metadata

Abstract: Grid layouts are used by designers to spatially organise user interfaces when sketching and wireframing. However, their design is largely time consuming manual work. This is challenging due to combinatorial explosion and complex objectives, such as alignment, balance, and expectations regarding positions. This paper proposes a novel optimisation approach for the generation of diverse grid-based layouts.

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4. Ngo, Teo & Byrne — Modelling interface aesthetics

Year: 2003 GitHub: No paper repository identified in REFERENCES.md/source metadata

Abstract: An important aspect of screen design is aesthetic evaluation of screen layouts. While it is conceivable to define a set of variables that characterize the key attributes of many alphanumeric display formats, such a task seems difficult for graphic displays because of their much greater complexity. This paper proposes a theoretical approach to capture the essence of artists' insights with 14 aesthetic measures for graphic displays.

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5. SlidesGen-Bench: Evaluating Slides Generation via Computational and Quantitative Metrics

Year: 2026 GitHub: <https://github.com/YunqiaoYang/SlidesGen-Bench>

Abstract: The rapid evolution of Large Language Models (LLMs) has fostered diverse paradigms for automated slide generation, ranging from code-driven layouts to image-centric synthesis. However, evaluating these heterogeneous systems remains challenging, as existing protocols often struggle to provide comparable scores across architectures or rely on uncalibrated judgments. In this paper, we introduce SlidesGen-Bench, a benchmark designed to evaluate slide generation through a lens of three core principles: universality, quantification, and reliability. First, to establish a unified evaluation framework, we ground our analysis in the visual domain, treating terminal outputs as renderings to remain agnostic to the underlying generation method. Second, we propose a computational approach that quantitatively assesses slides across three distinct dimensions - Content, Aesthetics, and Editability - offering reproducible metrics where prior works relied on subjective or reference-dependent proxies. Finally, to ensure high correlation with human preference, we construct the Slides-Align1.5k dataset, a human preference aligned dataset covering slides from nine mainstream generation systems across seven scenarios. Our experiments demonstrate that SlidesGen-Bench achieves a higher degree of alignment with human judgment than existing evaluation pipelines. Our code and data are available at <https://github.com/YunqiaoYang/SlidesGen-Bench>.

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6. Seeing Like a Designer Without One: A Study on Unsupervised Slide Quality Assessment via Designer Cue Augmentation

Year: 2025 GitHub: No paper repository identified in REFERENCES.md/source metadata

Abstract: We present an unsupervised slide-quality assessment pipeline that combines seven expert-inspired visual-design metrics (whitespace, colorfulness, edge density, brightness contrast, text density, color harmony, layout balance) with CLIP-ViT embeddings, using Isolation Forest-based anomaly scoring to evaluate presentation slides. Trained on 12k professional lecture slides and evaluated on six academic talks (115 slides), our method achieved Pearson correlations up to 0.83 with human visual-quality ratings-1.79x to 3.23x stronger than scores from leading vision-language models (ChatGPT o4-mini-high, ChatGPT o3, Claude Sonnet 4, Gemini 2.5 Pro). We demonstrate convergent validity with visual ratings, discriminant validity against speaker-delivery scores, and exploratory alignment with overall impressions. Our results show that augmenting low-level design cues with multimodal embeddings closely approximates audience perceptions of slide quality, enabling scalable, objective feedback in real time.

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7. PPTAgent: Generating and Evaluating Presentations Beyond Text-to-Slides

Year: 2025 **GitHub:** <https://github.com/icip-cas/PPTAgent>

Abstract: Automatically generating presentations from documents is a challenging task that requires balancing content quality, visual design, and structural coherence. Existing methods primarily focus on improving and evaluating the content quality in isolation, often overlooking visual design and structural coherence, which limits their practical applicability. To address these limitations, we propose PPTAgent, which comprehensively improves presentation generation through a two-stage, edit-based approach inspired by human workflows. PPTAgent first analyzes reference presentations to understand their structural patterns and content schemas, then drafts outlines and generates slides through code actions to ensure consistency and alignment. To comprehensively evaluate the quality of generated presentations, we further introduce PPTEval, an evaluation framework that assesses presentations across three dimensions: Content, Design, and Coherence. Experiments show that PPTAgent significantly outperforms traditional automatic presentation generation methods across all three dimensions. The code and data are available at <https://github.com/icip-cas/PPTAgent>.

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8. DECKBench: Benchmarking Multi-Agent Frameworks for Academic Slide Generation and Editing

Year: 2026 **GitHub:** <https://github.com/morgan-heisler/DeckBench>

Abstract: Automatically generating and iteratively editing academic slide decks requires more than document summarization. It demands faithful content selection, coherent slide organization, layout-aware rendering, and robust multi-turn instruction following. However, existing benchmarks and evaluation protocols do not adequately measure these challenges. To address this gap, we introduce the Deck Edits and Compliance Kit Benchmark (DECKBench), an evaluation framework for multi-agent slide generation and editing. DECKBench is built on a curated dataset of paper to slide pairs augmented with realistic, simulated editing instructions. Our evaluation protocol systematically assesses slide-level and deck-level fidelity, coherence, layout quality, and multi-turn instruction following. We further implement a modular multi-agent baseline system that decomposes the slide generation and editing task into paper parsing and summarization, slide planning, HTML creation, and iterative editing. Experimental results demonstrate that the proposed benchmark highlights strengths, exposes failure modes, and provides actionable insights for improving multi-agent slide generation and editing systems. Overall, this work establishes a standardized foundation for reproducible and comparable evaluation of academic presentation generation and editing. Code and data are publicly available at <https://github.com/morgan-heisler/DeckBench>.

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9. G. D. Birkhoff — Aesthetic Measure

Year: 1933 **GitHub:** No paper repository identified in REFERENCES.md/source metadata

Abstract: This article has no associated abstract.

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10. Harrington et al. — Aesthetic measures for automated document layout

Year: 2004 **GitHub:** No paper repository identified in REFERENCES.md/source metadata

Abstract: A measure of aesthetics that has been used in automated layout is described. The approach combines heuristic measures of attributes that degrade the aesthetic quality. The combination is nonlinear so that one bad aesthetic feature can harm the overall score. Example heuristic measures are described for the features of alignment regularity separation balance white-space fraction white-space free flow proportion uniformity and page security.

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11. Rebelo, Merelo, Bicker & Machado — Evaluation Metrics for Automated Typographic Poster Generation

Year: 2024 **GitHub:** No paper repository identified in REFERENCES.md/source metadata

Abstract: Computational Design approaches facilitate the generation of typographic design, but evaluating these designs remains a challenging task. In this paper, we propose a set of heuristic metrics for typographic design evaluation, focusing on their legibility, which assesses the text visibility, aesthetics, which evaluates the visual quality of the design, and semantic features, which estimate how effectively the design conveys the content semantics. We experiment with a constrained evolutionary approach for generating typographic posters, incorporating the proposed evaluation metrics with varied setups, and treating the legibility metrics as constraints. We also integrate emotion recognition to identify text semantics automatically and analyse the performance of the approach and the visual characteristics outputs.

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12. Reinecke et al. — Predicting Users' First Impressions of Website Aesthetics with a Quantification of Perceived Visual Complexity and Colorfulness

Year: 2013 **GitHub:** No paper repository identified in REFERENCES.md/source metadata

Abstract: Users make lasting judgments about a website's appeal within a split second of seeing it for the first time. This first impression is influential enough to later affect their opinions of a site's usability and trustworthiness. In this paper, we demonstrate a means to predict the initial impression of aesthetics based on perceptual models of a website's colorfulness and visual complexity. In an online study, we collected

ratings of 450 websites by 548 volunteers, and found that our models explain almost half of the variance in the ratings of the first impressions of appeal.

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13. Kikuchi et al. — Constrained Graphic Layout Generation via Latent Optimization

Year: 2021 GitHub: https://github.com/ktrk115/const_layout

Abstract: It is common in graphic design humans visually arrange various elements according to their design intent and semantics. For example, a title text almost always appears on top of other elements in a document. In this work, we generate graphic layouts that can flexibly incorporate such design semantics, either specified implicitly or explicitly by a user. We optimize using the latent space of an off-the-shelf layout generation model, allowing our approach to be complementary to and used with existing layout generation models. Our approach builds on a generative layout model based on a Transformer architecture, and formulates the layout generation as a constrained optimization problem where design constraints are used for element alignment, overlap avoidance, or any other user-specified relationship. We show in the experiments that our approach is capable of generating realistic layouts in both constrained and unconstrained generation tasks with a single model. The code is available at https://github.com/ktrk115/const_layout.

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14. O'Donovan, Agarwala & Hertzmann — Learning Layouts for Single-Page Graphic Designs

Year: 2014 GitHub: No paper repository identified in REFERENCES.md/source metadata

Abstract: This paper presents an approach for automatically creating graphic design layouts using a new energy-based model derived from design principles. The model includes several new algorithms for analyzing graphic designs, including the prediction of perceived importance, alignment detection, and hierarchical segmentation. Given the model, we use optimization to synthesize new layouts for a variety of single-page graphic designs.

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15. Miniukovich & De Angeli — Computation of Interface Aesthetics

Year: 2015 GitHub: No paper repository identified in REFERENCES.md/source metadata

Abstract: People prefer attractive interfaces. Designers strive to outmatch competitors, and create apps and websites that stand out. However, significant expenses on design are unaffordable to small companies; instead, they could adopt automatic tools of interface aesthetics evaluation, a cheaper strategy to good design. This paper describes an important step towards such a tool; it presents eight

automatic metrics of graphical user interface (GUI) aesthetics.

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